Karthik Bharadwaj

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Summary

An AI enthusiast and Software engineer who likes building things that scale and breaking things that don't. 2 years of shipping backend systems, APIs, and cloud infrastructure. Currently chasing a Master's in Computer Science at Northeastern (4.0 GPA) and tinkering with AI on the side. Open to Summer 2026 internships and Fall 2026 co-ops.

Experience

Digitide Solutions Ltd — **Sep 2024 – Jun 2025**
Digital Associate – Software Engineer — Bangalore, India

- Developed an NLP-powered chatbot that integrated AI models with backend services, enabling warehouse staff to query inventory via natural language and reducing manual lookup time by 60%.
- Delivered RESTful API-driven SaaS modules for AB-InBev serving 500+ daily users, including report generation, admin dashboards, and bulk data workflows.
- Architected and deployed backend services on Azure using Docker containers, reducing release turnaround time by 2 days.

zevvo — **Jan 2024 – Jul 2024**
Software Engineer – Full Stack — Bangalore, India

- Built scalable backend systems from scratch for bookings, billing, and catalog management with MongoDB schema design, JWT authentication, and RESTful API conventions, onboarding 50+ active users within the first quarter.
- Integrated third-party payment and notification SDKs into the backend service layer, enabling real-time transaction processing and reducing checkout drop-off by 15%.
- Streamlined CI/CD via GitHub Actions, deploying Dockerized services to AWS (ECS, S3, ECR) with auto-scaling and increasing deploy frequency by 40%.

Crestron Electronics — **Mar 2023 – Jul 2023**
Software Development Intern — Bangalore, India

- Optimized .NET Core REST APIs with rewritten SQL queries, cutting average response time by 28% across production endpoints.
- Elevated unit test coverage from 65% to 89% through Jest suites, reducing post-deployment bugs and shortening the QA feedback loop.

Education

Northeastern University — **Expected Dec 2027**
Master of Science in Computer Science; GPA: 4.0/4.0 — Boston, MA

- Coursework: Algorithms, DBMS, Foundations of AI, Programming Design Paradigms, Machine Learning, NLP

R V College of Engineering — **May 2023**
Bachelor of Engineering in Computer Science and Engineering; CGPA: 8.04/10 — Bangalore, India

Skills

- **Languages:** Python, SQL, Java, TypeScript, JavaScript, R, C++, C#
- **Backend:** FastAPI, Node.js, .NET Core, Express.js, SQLAlchemy, REST APIs, Microservices, Event-Driven Architecture
- **Frontend:** React.js, Next.js, Angular, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Firebase
- **Cloud/DevOps:** AWS (S3, ECS, Lambda, EC2), GCP, Azure, Docker, Kubernetes, GitHub Actions, CI/CD
- **ML/AI:** PyTorch, scikit-learn, OpenCV, YOLOv3, CNNs, Transfer Learning, NLP, AI Model Integration
- **Practices:** API Design, System Design, Distributed Systems, Caching, OOP, Agile, TDD, Git

Projects

ParetoFolio – Portfolio Optimization Engine Python, FastAPI, Pydantic v2, React 19, TypeScript

- Constructed a FastAPI backend with Pydantic v2 request validation, yfinance data ingestion, and a covariance matrix computation pipeline, serving a typed RESTful API consumed by a React 19 frontend.
- Implemented NSGA-II entirely from scratch in Python, including SBX crossover, polynomial mutation, tournament selection, and crowding distance, removing dependency on external optimization libraries.
- Designed a two-objective optimization strategy for portfolio construction, producing a ranked set of Pareto-optimal portfolios across 75 US equities with an interactive efficient frontier plot and Sharpe-ranked table.

SportsTV Streaming Data Warehouse R, MySQL 8, SQLite, R Markdown, Kimball Star Schema

- Engineered a bulk-batched, idempotent ETL data pipeline in R with surrogate-key resolution and partition pruning, achieving a 99%+ source-to-warehouse match rate across 2M+ streaming transactions.
- Modeled a Kimball-style star schema with five conformed dimensions and a RANGE-partitioned fact table on cloud-hosted MySQL 8, consolidating heterogeneous data sources into a unified analytical warehouse.
- Generated a fully automated BI report with KPI summaries, growth trends, and device breakdowns, driven entirely by live SQL queries with zero hard-coded values.

Autonomous Vehicle Perception Pipeline Python, PyTorch, YOLOv3, OpenCV, NumPy

- Fine-tuned YOLOv3 on a custom traffic sign dataset, achieving 88% mAP for real-time detection across varying lighting and weather conditions.
- Created a multi-sensor fusion pipeline combining camera and distance inputs for autonomous steering and braking, enabling lane-keeping without manual waypoint programming.